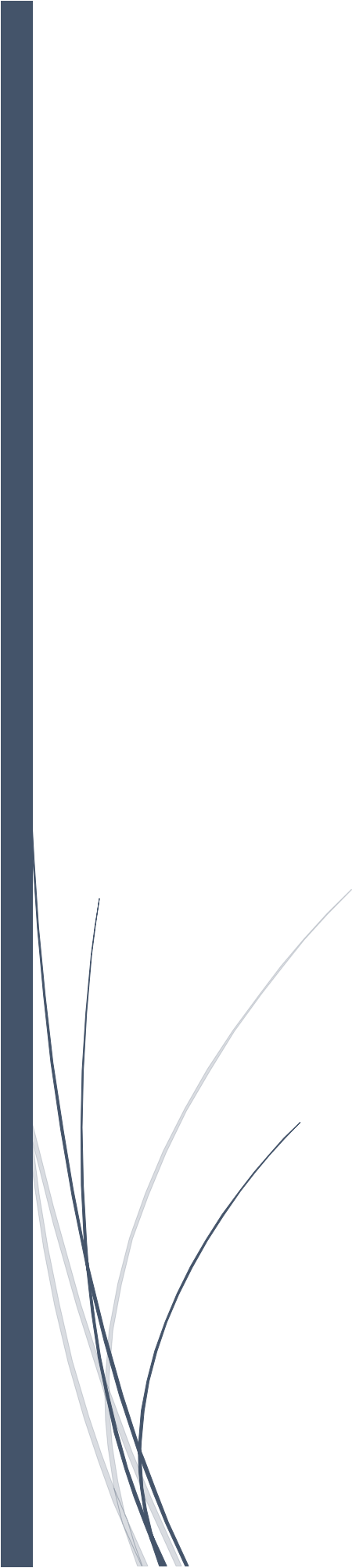


A thick dark blue vertical bar is positioned on the left side of the page. From its base, several thin, curved lines in dark blue and light grey extend upwards and outwards, creating an abstract, organic shape.

Varistor selection Guideline

How to use the sheet

Abdallah Ebraheem Abdelmotalb Keshk

INPUTS		
Input Parameters	Value	Note
Applied Surge Test Voltage		From IEC--61000-4-5 *Set the worst as 2KV* IN V
Input Voltage (AC)		AC main Input in V
Maximum Power dissipation of the varistor		FROM DATASHEET IN WATT
maximum possible energy absorption		In Joules
Varistor Voltage @ the short circuit conditions		From datasheet of the Varistor in V/I Curve at the short circuit conditons in V (see the pages)
The clamping voltage of the Varistor at the actual of current in the varistor		From datasheet of the Varistor in V/I Curve at actual current flowing through varistor in V (see the pages)
Maximum allowed clamping voltage		(Dependes on the reverse voltage of the rectifier usually) in V EX (14007 RECTIFIRE IS 1000V reverse voltage) OR MAXIMUM REQUIRED V CLAMP OF THE SYSTEM

1. Applied surge test worst case is +-2KV According to IEC-61004-5
2. Input voltage of the AC which is usually 220 +-10%
3. Maximum power dissipation of the varistor mentioned in the datasheet as below for
TVR10431KSY

Electrical Characteristics (Ambient $T_a=25^{\circ}\text{C}$)

Part No.	Varistor Voltage (@ 1mA DC)	Max. Continuous Voltage		Max. Clamping Voltage (8/20 μ S)		Max. Surge Current (8/20 μ S)	Max. Energy (10/1000 μ S)
	V_{1mA} (V)	$V_{AC(rms)}$ (V)	V_{DC} (V)	V_p (V)	I_p (A)	I (A)	W (J)
TVR10431KSY	430 $\pm$ 10 %	275	350	710	25	2500	65

Part No.	Rated Power	Impulse Response Time	Max. Leakage Current at 75% V_{1mA}	Operating Temperature Range	Storage temperature Range
	P (W)	nSec	$I_L(\mu A)$	($^{\circ}\text{C}$)	($^{\circ}\text{C}$)
TVR10431KSY	0.4	<25	20	-40 +105	-40 ~ +125

PD IS 0.4

Max Energy can be absorbed is 65 J

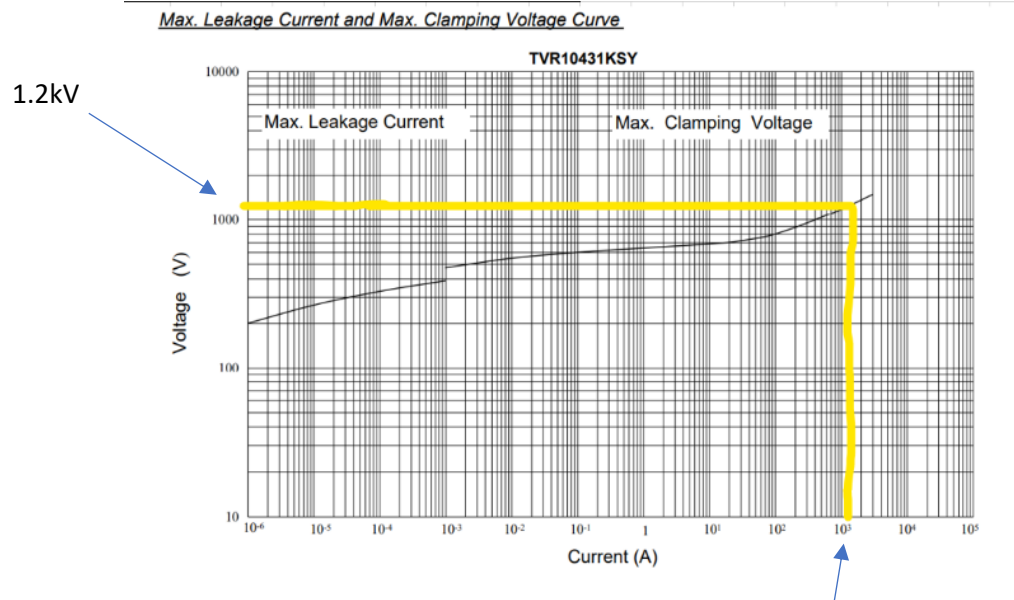
5- The clamping voltage of the Varistor at the actual of current in the varistor

But the value extracted above in the table as shown

	A	B	C
1	INPUTS		
2	Input Parameters	Value	Note
3	Applied Surge Test Voltage	2000	From IEC--61000-4-5 *Set the worst as 2KV* IN V
4	Input Voltage (AC)	220	AC main Input in V
5	Maximum Power dissipation of the varistor	0.4	FROM DATASHEET IN WATT
5	maximum possible energy absorption	65	In Joules
7	Varistor Voltage @ the short circuit conditions		From datasheet of the Varistor in V/I Curve at the short circuit conditons in V (see the pages)
3	The clamping voltage of the Varistor at the actual of current in the varistor		From datasheet of the Varistor in V/I Curve at actual current flowing through varistor in V (see the pages)
9	Maximum allowed clamping voltage		(Dependes on the reverse voltage of the rectifier usually) in V EX (14007 RECTIFIRE IS 1000V reverse voltage) OR MAXIMUM REQUIRED V CLAMP OF THE SYSTEM
0			
1			
2	Automatic Equations caculation		
3	Maximum Short Circuit current (Automatic calculated)	1155.54	(NOT input leave the equation alone) IN A
4	Actual Current in the Varistor	1155.54	calaculated in A
5	The Maximum Energy from the pulses	0	calaculated in A in JULE

The automated short circuit equation results I **1155.4 A**

Go to the V/I Curve and extract the clamping voltage @ the current calculated above



The Extracted Clamp voltage is **1.2KV** @ short circuit condition **1155.4A**

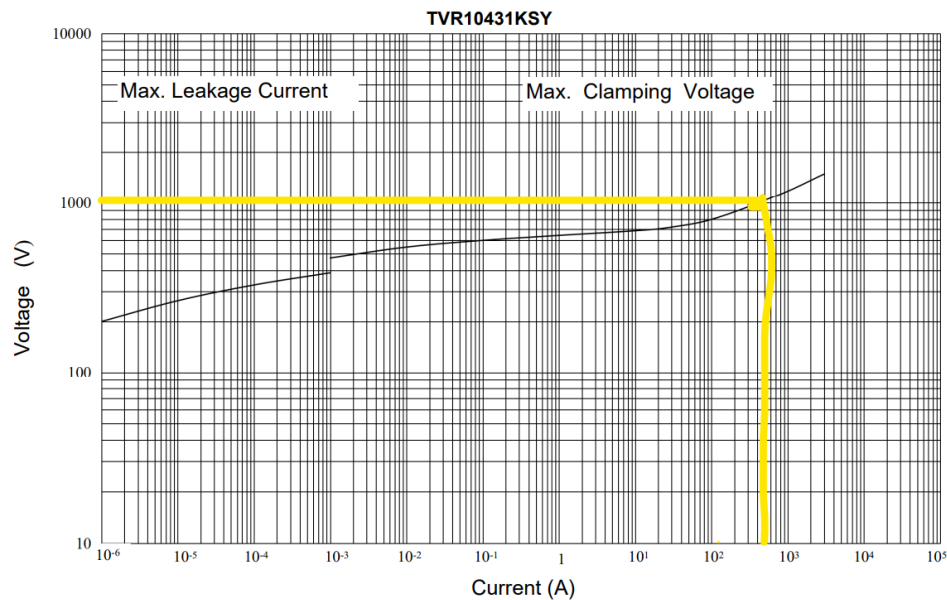
Put this value in the Varistor Voltage @ the short circuit conditions as shown

INPUTS		
Input Parameters	Value	Note
Applied Surge Test Voltage	2000	From IEC--61000-4-5 *Set the worst as 2KV* IN V
Input Voltage (AC)	220	AC main Input in V
Maximum Power dissipation of the varistor	0.4	FROM DATASHEET IN WATT
maximum possible energy absorption	65	In Joules
Varistor Voltage @ the short circuit conditions	1200	From datasheet of the Varistor in V/I Curve at the short circuit conditons in V (see the pages)
The clamping voltage of the Varistor at the actual of current in the varistor		From datasheet of the Varistor in V/I Curve at actual current flowing through varistor in V (see the pages)
Maximum allowed clamping voltage		(Dependes on the reverse voltage of the rectifier usually) in V EX (14007 RECTIFIRE IS 1000V reverse voltage) OR MAXIMUM REQUIRED V CLAMP OF THE SYSTEM
Automatic Equations calaculation		
Maximum Short Circuit current (Automatic calculated)	1155.54	(NOT input leave the equation alone) IN A
Actual Current in the Varistor	555.54	calaculated in A

You will notice the new value of actual current in the varistor which is **555.54 A**

Then extract the value of clamp voltage again using the actual varistor current

Max. Leakage Current and Max. Clamping Voltage Curve



The extracted clamp voltage is **1 KV**

4. Put this value again in the table as shown

INPUTS		
Input Parameters	Value	Note
Applied Surge Test Voltage	2000	From IEC--61000-4-5 *Set the worst as 2KV* IN V
Input Voltage (AC)	220	AC main Input in V
Maximum Power dissipation of the varistor	0.4	FROM DATASHEET IN WATT
maximum possible energy absorption	65	In Joules
Varistor Voltage @ the short circuit conditions	1200	From datasheet of the Varistor in V/I Curve at the short circuit conditons in V (see the pages)
he clamping voltage of the Varistor at the actual of current in the varistor	1000	From datasheet of the Varistor in V/I Curve at actual current flowing through varistor in V (see the pages)
Maximum allowed clamping voltage		(Dependes on the reverse voltage of the rectifier usually) in V EX (14007 RECTIFIRE IS 1000V reverse voltage) OR MAXIMUM REQUIRED V CLAMP OF THE SYSTEM

5. Put the maximum allowed clamping voltage is the maximum, voltage can be applied at the rest of the circuit which is typically 1000V for the reverse voltage of the rectifier diode for 1n4007 Rectifier diode

INPUTS		
Input Parameters	Value	Note
Applied Surge Test Voltage	2000	From IEC--61000-4-5 *Set the worst as 2KV* IN V
Input Voltage (AC)	220	AC main Input in V
Maximum Power dissipation of the varistor	0.4	FROM DATASHEET IN WATT
maximum possible energy absorption	65	In Joules
Varistor Voltage @ the short circuit conditions	1200	From datasheet of the Varistor in V/I Curve at the short circuit conditons in V (see the pages)
The clamping voltage of the Varistor at the actual of current in the varistor	1000	From datasheet of the Varistor in V/I Curve at actual current flowing through varistor in V (see the pages)
Maximum allowed clamping voltage	1000	(Dependes on the reverse voltage of the rectifier usually) in V EX (14007 RECTIFIRE IS 1000V reverse voltage) OR MAXIMUM REQUIRED V CLAMP OF THE SYSTEM

You are now done

Automatic Equations calculation		
Maximum Short Circuit current (Automatic calculated)	1155.54	(NOT input leave the equation alone) IN A
Actual Current in the Varistor	555.54	calaculated in A
The Maximum Energy from the pulses	9.40418112	calaculated in A in JULE
Automatic Checklist of the selection		
does the varistor have the enough time to cool	YES	Safe to use
does the varistor have the ability to abosrb the surge energy	YES	Safe to use
does the varistor Complies with the Required max clamp voltage needed	NO	Not safe to use

As for the result it is not recommended to use because the clamping voltage during the test is equal to the maximum voltage allowed for the rest of the circuit which is 1000 V if it is 1n4007

But for other factors it's OK